



JanDrishti — Architecture & Governance Miro
Canvas

Problem Statement ID: SIH26102 | Parliamentary Intelligence & Statutory Audit Platform

REACT 19 +
FASTAPI

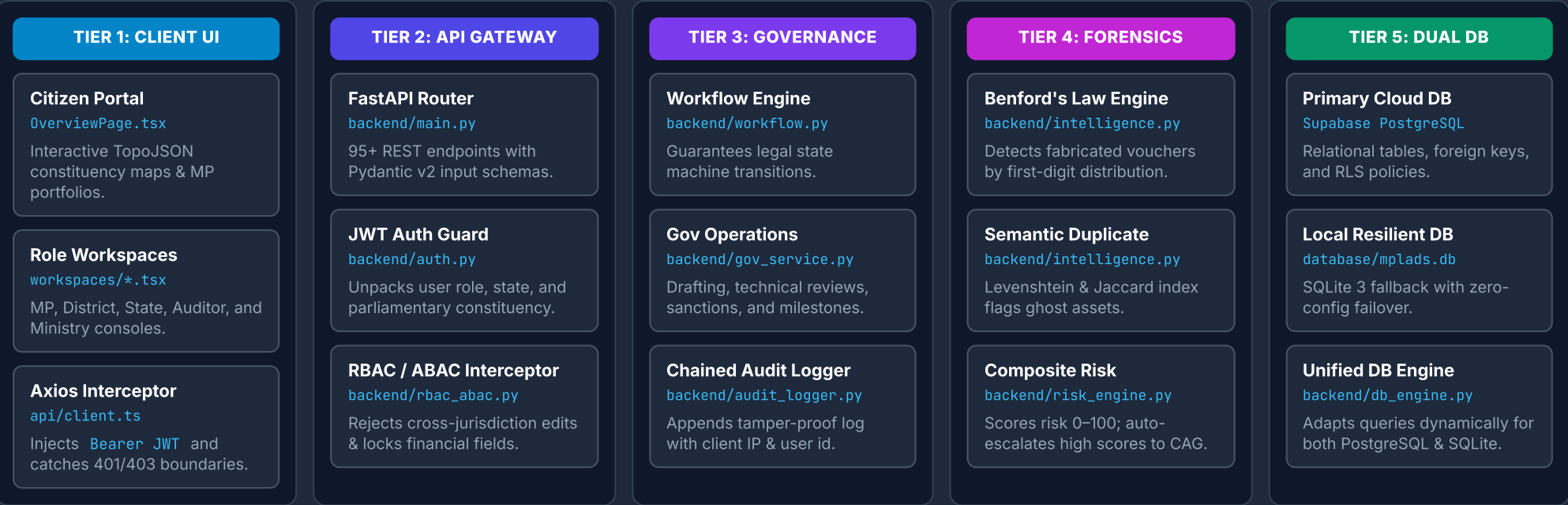
100% ZERO-MOCK
DATA

6 STATUTORY
RANKS

PRE-DISBURSEMENT
FORENSICS

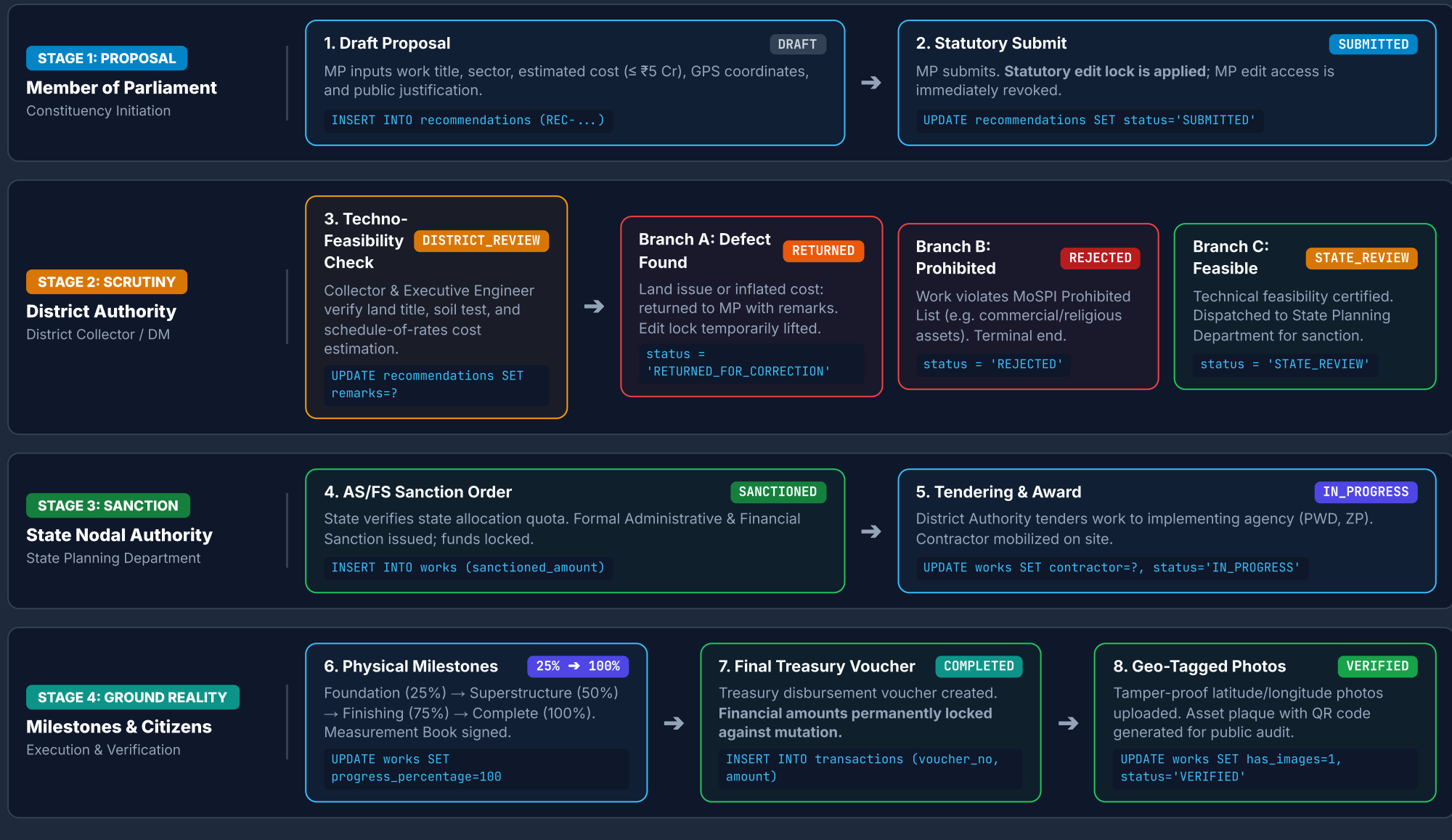
 Board 1: Full System Data Flow & Technology Architecture

Complete 5-Tier layout fitting in a single view: Where user requests originate, how security intercepts them, and where records persist.



Board 2: Statutory Governance Lifecycle (From MP Idea to Ground Verification)

Step-by-step pipeline showing what actually happens, who decides, what database tables change, and all branch conditions.



Board 3: Pre-Disbursement Forensic Audit Intelligence & Fraud Screening

Parallel statistical screening before public money is disbursed, calculating a composite risk score to prevent fiscal fraud.

1. Ingested Transaction

Whenever a work voucher is recorded or updated, it passes through 5 mathematical screening engines in real time.

Voucher Amount: ₹24,80,000
Contractor: M/S Apex Infratech
Location: Ward 4, Block X
Timestamp: 28-MAR-2026

1. Benford's Law

$P(d) = \log_{10}(1 + 1/d)$ (Chi-sq p < 0.05)

Weight: 20%

2. Semantic Duplication

Jaccard > 0.85 & Levenshtein < 3

Weight: 35%

3. Vendor Cartel (HHI)

$HHI = \sum (\text{Market Share})^2$ (> 2500)

Weight: 25%

4. Velocity Spikes

Z-Score > 3.0 (March-rush spike)

Weight: 20%

5. Predictive Delay

RandomForestRegressor(District Track)

Auxiliary

Composite Risk Action

🚨 Score ≥ 75 (High Risk)

Auto-escalated to CAG Auditor Docket in [review_cases](#) ; Real-time Alert sent to Ministry; Payment lock recommended.

⚠️ Score 45–74 (Medium Risk)

Advisory flag placed on District Collector's dashboard; Executive Engineer explanation required.

✅ Score < 45 (Low Risk)

Normal treasury clearance granted.

⚡ Board 4: Full Code Request-Response Lifecycle (Front-to-Back)

Step-by-step technical execution: How a user interaction in React 19 travels through security gating to database persistence.

1. React 19 UI

MpWorkspace.tsx

MP clicks "Submit Recommendation". Event handler dispatches payload to Axios client.

api/client.ts

Attaches `Authorization: Bearer <JWT>` and posts to `/api/v1/recommendations/submit`.

Optimistic UI

Sets local button loading state to prevent duplicate rapid submissions.

Error Boundary

Catches 401/403 errors and displays contextual statutory error toast.

Cache Invalidation

On 200 OK, invalidates React Query cache and badges status as "SUBMITTED".

2. Security Gate

backend/auth.py

Decodes HMAC-SHA256 JWT; extracts `AuthenticatedUser` (role='MP', state='MAHARASHTRA').

backend/rbac_abac.py

Verifies MP role has 'SUBMIT' permission and record state matches user's territorial boundary.

Immutability Lock

Verifies request does not tamper with `sanctioned_amount` or primary keys.

backend/scope.py

Generates mandatory SQL fragment: `WHERE state_normalized = ? AND internal_mp_id = ?`.

Boundary Check

Rejects cross-jurisdiction requests with HTTP 403 Forbidden.

3. Backend Engine

backend/workflow.py

Verifies transition from 'DRAFT' → 'SUBMITTED' is legally permitted for 'ROLE_MP'.

gov_service.py

Updates recommendation state and records district remarks.

db_engine.py

Executes parameterized UPDATE against Supabase Cloud PostgreSQL.

Failover Driver

If cloud connection drops, automatically executes against local SQLite fallback.

audit_logger.py

Inserts tamper-proof record into `audit_logs` with IP, user id, and timestamp.

🛡️ Board 5: Statutory Authority Matrix (Separation of Powers)

Clear demarcation of powers and territorial boundaries across Indian public finance tiers.

Role & Rank	Jurisdiction Scope	Statutory Powers (What they CAN do)	Strict Restrictions (What they CANNOT do)
RANK 1 MINISTRY_ADMIN	Pan-India (National Corpus)	View all 36 States, configure system risk weights, approve central dockets, manage system users, review national audit alerts.	Cannot modify ground physical milestone records; cannot bypass district techno-economic feasibility.
RANK 2 STATE_NODAL	Designated State (<code>user.state</code>)	Review district dockets, grant statutory Financial Sanction (AS/FS), reject non-compliant works, manage state budget quotas.	Cannot access, approve, or modify records of any other Indian State.
RANK 3 DISTRICT_AUTHORITY	Designated District (<code>user.district</code>)	Techno-economic scrutiny, tender floating, contractor assignment, milestone updates, geo-tagged photo uploads, release final payment.	Cannot recommend works; cannot sanction outside district; cannot modify treasury transaction values directly.

Role & Rank	Jurisdiction Scope	Statutory Powers (What they CAN do)	Strict Restrictions (What they CANNOT do)
RANK 4 MEMBER OF PARLIAMENT	Parliamentary Seat (<code>user.mp_id</code>)	Draft and submit work recommendations within ₹5 Crore entitlement; track execution velocity; view constituency ledger.	Locked from editing after submission; cannot approve works; cannot access other MPs' dockets.
RANK 5 STATUTORY AUDITOR	National Mandate (CAG / Independent)	Inspect Benford flags, cartel HHI scores, duplicate work dockets; create forensic audit cases; flag transactions for recovery.	Read-only to financial ledgers; cannot disburse, approve, or alter financial figures.
RANK 6 GENERAL CITIZEN	Open Public (National Read-Only)	Inspect all works on interactive maps; search MP utilization; scan QR code plaques on physical assets; submit ground discrepancy feedback.	Zero mutation power on official records; cannot edit or sanction works.

🔒 Absolute Financial Immutability Policy (ABAC)

Under `backend/rbac_abac.py`, fields including `sanctioned_amount`, `expenditure_amount`, `work_id`, and `internal_transaction_id` strictly reject generic HTTP PUT/PATCH updates. They can only be updated via verified Treasury Voucher reconciliation ledger operations.